1) Consider the velocity field given, by Lecture 4

$$u = \frac{y}{(x^2+y^2)} \qquad v = \frac{-x}{x^2+y^2}$$

Calculate the equation of the streamline passing through the point $(0, 5)$

$$\frac{dy}{dx} = \frac{v}{u} = -\frac{x}{y}$$

$$\int y\, dy = \int -x\, dy$$

$$y^2 = -x^2 + C$$

$$y^2 + x^2 = 25$$

$$\psi = \int u\, dy \longrightarrow \psi = \frac{1}{2}\ln(x^2+y^2) + \text{Const.}$$

$$C = \frac{1}{2}\ln(x^2+y^2) + \text{Const}$$

$$e^{2(C-\text{const})} = x^2+y^2$$

Lecture 2

2) For the velocity given by $u = \frac{y}{x^2+y^2}$ $v = \frac{-x}{x^2+y^2}$

Calculate the vorticity

$$\xi_z = \vec{\nabla} \times v = \hat{k}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

$$\frac{\partial v}{\partial x} = \frac{-1(x^2+y^2) + x(2x)}{(x^2+y^2)}$$

$$\frac{\partial u}{\partial x} = \frac{(x^2+y^2) - y(2y)}{(x^2+y^2)^2}$$

$$= 0$$

Lecture 2

3) For velocity field $u = \frac{y}{x^2+y^2}$ $v = \frac{-x}{(x^2+y^2)}$ 1

calculate Γ around a circular path wl radius = 5m

$x = r\cos\theta$

$y = r\sin\theta$

$x^2+y^2 = r$

$V_r = u\cos\theta + v\sin\theta$

$V_\theta = -u\sin\theta + v\cos\theta$

$$\Gamma = -\oint_c \vec{V}\cdot d\vec{s}$$

$$u = \frac{y}{x^2+y^2} = \frac{r\sin\theta}{r^2} = \frac{\sin\theta}{r}$$

$$v = \frac{-x}{x^2+y^2} = \frac{-r\cos\theta}{r^2} = -\frac{\cos\theta}{r}$$

$$V_r = \frac{\sin\theta}{r}\cos\theta - \frac{\cos\theta}{r}\sin\theta = 0$$

$$V_\theta = -\frac{\sin\theta}{r}\sin\theta + -\frac{\cos\theta}{r}\cos\theta$$

$$-\frac{(\sin^2\theta + \cos^2\theta)}{r} = -\frac{1}{r}$$

$$\vec{V}\cdot d\vec{s} = (V_r\hat{e}_r + V_\theta\hat{e}_\theta)\cdot(dr\,\hat{e}_r + r\,d\theta\,\hat{e}_\theta)$$

$$\nabla_r \cdot d_r + \nabla_\theta \cdot d_\theta = D + r\left(-\frac{1}{r}\right) d_\theta = -d_\theta$$

$$-\int_{-\partial\pi}^{\partial\pi} -d_\theta = \boxed{-\partial\pi}$$

Lecture 2

4) $\vec{V} = (3xy^2 - x^3)\,\hat{i} + (3x^2y - y^3)\,\hat{j}$

a) Is this flow physically possible?

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\frac{\partial}{\partial x}(3xy^2 - x^3) + \frac{\partial}{\partial y}(3x^2y - y^3) = 0\ ?$$

$$3y^2 - 3x^2 + 3x^2 - 3y^2 = 0 \quad \checkmark$$

b) If so, what is the volumetric flow rate (Q) between 2 streamlines that pass through (0,0) & (1,3)?

$$u = \frac{\partial \psi}{\partial y} \qquad v = -\frac{\partial \psi}{\partial x}$$

$$3xy^2 - x^3 = \frac{\partial \psi}{\partial y} \rightarrow \psi = xy^3 - yx^3 + f(x)$$

$$3x^2y - y^3 = -\frac{\partial}{\partial x}(xy^3 - yx^3 + f(x))$$

$$3x^2y - y^3 = -[y^3 - 3yx^2 + f'(x)]$$

$$0 = f'(x)$$

$$C = f(x)$$

$$\boxed{\psi = xy^3 - yx^3 + C}$$

Assume $\psi(x,y) = A$

$$A = xy^3 - yx^3 + C \rightarrow S = xy^3 - x^3y \quad \text{where} \quad S = A - C$$

$$Q = S_2 - S_1$$

$$S_2 = (1,3) \qquad S_1 = (0,0)$$

$$Q = 24 - 0 = 24$$

C) Find the velocity potential

$$u = \frac{\partial \phi}{\partial x} \qquad v = \frac{\partial \phi}{\partial y}$$

$$3xy^2 - x^3 = \frac{\partial \phi}{\partial x} \rightarrow \phi = \frac{3}{2}x^2y^2 - \frac{x^4}{4} + f(y)$$

$$3x^2y - y^3 = \frac{\partial}{\partial y}\left(\frac{3}{2}x^2y^2 - \frac{x^4}{4} + f(y)\right)$$

$$\cancel{3x^2y} - y^3 = \cancel{3x^2y} + f'(y)$$

$$f'(y) = -y^3 \rightarrow f(y) = -\frac{y^4}{4} + C$$

$$\boxed{\phi = \frac{3}{2}x^2y^2 - \frac{x^4}{4} - \frac{y^4}{4} + C}$$

Lecture 3

5) Consider an airfoil in a flow at a standard sea level conditions w/ a freestream velocity of 50 m/s. At a given point on the airfoil, the pressure is 0.9×10^5 N/m. Calculate the velocity at this point.

Standard sea level conditions:

$\rho_\infty = 1.225 \ kg/m^3$

$P_\infty = 1.01 \times 10^5 \ N/m^3$

$V_\infty = 50 \ m/s$

$P = 0.9 \times 10^5 \ P_a$

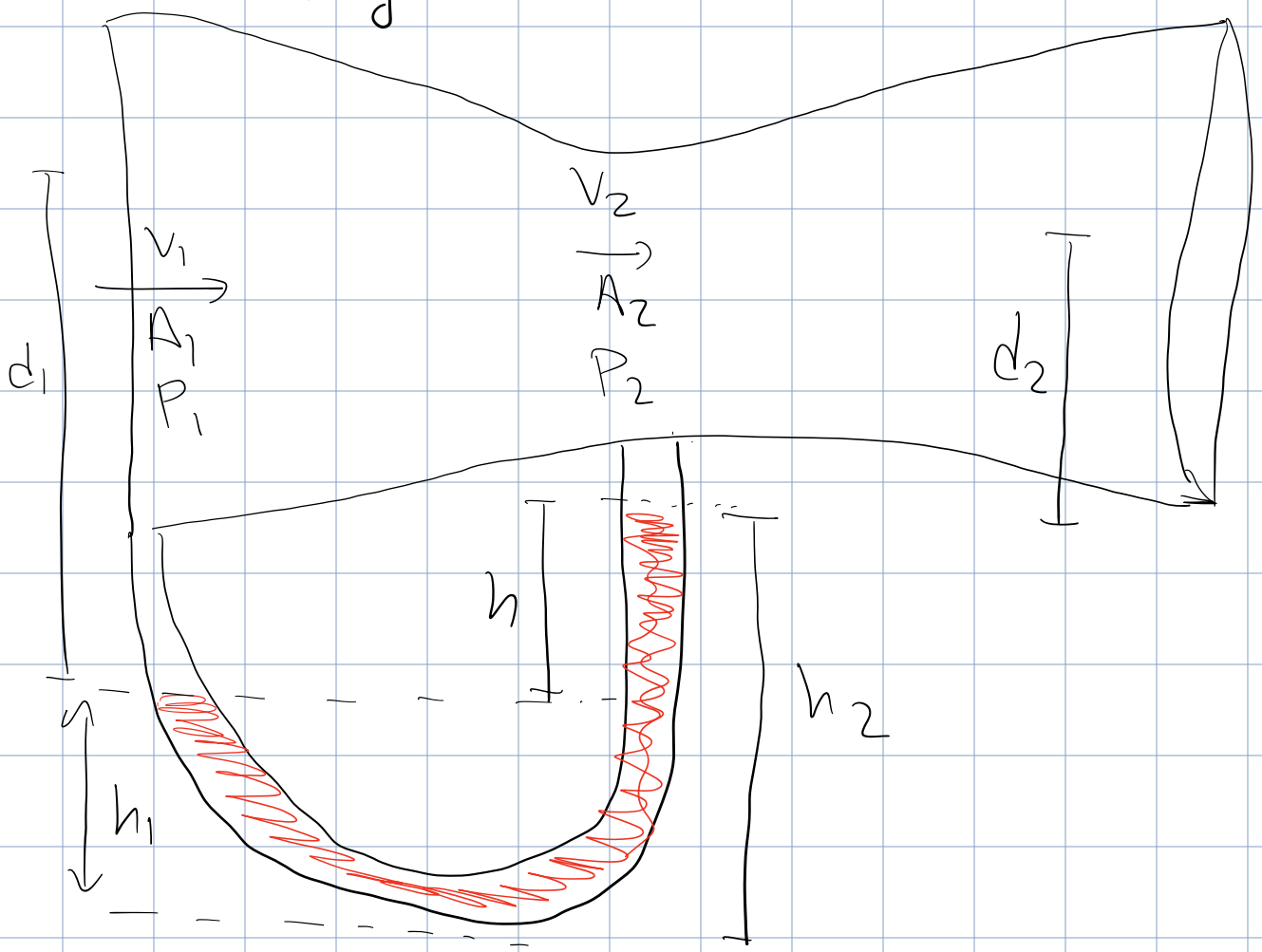
$V = ?$

$$P_\infty + \frac{1}{2}\rho V_\infty^2 = P + \frac{1}{2}\rho V^2$$

$$V = \sqrt{\frac{2(P_\infty - P)}{\rho} + V_\infty^2} = \boxed{142.8 \ m/s}$$

Lecture 3

6) A venturi meter → Device that can be used to measure the speed of a fluid in a pipe. A fluid of density ρ_f is flowing through a pipe. A U-shaped tube partially filled w/ mercury of density ρ_{Hg} lies underneath the points 1 & 2.



Bernoullis:

$$P_1 + \frac{1}{2}\rho_f v_1^2 = P_2 + \frac{1}{2}\rho_f v_2^2$$

$$P_1 - P_2 = \frac{1}{2}\rho_f (v_2^2 - v_1^2)$$

Hydrostatics

$$P_1 + \rho_F g d_1 + \rho_{Hg} g h_1 = P_2 + \rho_F g d_2 + \rho_{Hg} g h_2$$

$$d_1 + h_2 = d_1 + h_1 \longrightarrow d_2 - d_1 = h_2 - h_1 = h$$

$$P_1 - P_2 = (\rho_{Hg} - \rho_F) g h$$

$$\frac{1}{2} \rho_F (v_2^2 - v_1^2) = (\rho_{Hg} - \rho_F) g h$$

$$A_1 v_1 = A_2 v_2$$

$$v_2 = \frac{A_1}{A_2} v_1$$

$$\frac{1}{2} \rho_F \left[\left(\frac{A_1}{A_2} \right) v_1^2 - v_1^2 \right] = (\rho_{Hg} - \rho_F) g h$$

$$\frac{1}{2} \rho_F \left[\left(\frac{A_1}{A_2} \right) - 1 \right] v_1^2 = (\rho_{Hg} - \rho_F) g h$$

$$\boxed{v = \sqrt{\frac{2(\rho_{Hg} - \rho_F) g h}{\rho_F \left[\left(\frac{A_1}{A_2} \right)^2 - 1 \right]}}}$$